

Experiment No-5

Generation of signals and sequences

AIM: Generate various signals and sequences (Periodic and aperiodic), such as Unit Impulse, Unit Step, Square, Saw tooth, Triangular, Sinusoidal, Ramp, Sinc.

Software Required: Matlab software

Theory: If the amplitude of the signal is defined at every instant of time then it is called continuous time signal. If the amplitude of the signal is defined at only at some instants of time then it is called discrete time signal. If the signal repeats itself at regular intervals then it is called periodic signal. Otherwise they are called aperiodic signals.

EX: ramp, Impulse, unit step, sinc- Aperiodic signals
square, sawtooth, triangular sinusoidal – periodic signals.

Ramp signal: The **ramp function** is a unitary real function, easily computable as the mean of the independent variable and its absolute value. This function is applied in engineering. The name *ramp function* is derived from the appearance of its graph.

$$r(t) = \begin{cases} t & \text{when } t \geq 0 \\ 0 & \text{else} \end{cases}$$

Unit impulse signal: One of the more useful functions in the study of linear systems is the "unit impulse function". An ideal impulse function is a function that is zero everywhere but at the origin, where it is infinitely high. However, the *area* of the impulse is finite

$$Y(t) = \begin{cases} 1 & \text{when } t=0 \\ 0 & \text{other wise} \end{cases}$$

Unit step signal: The unit step function and the impulse function are considered to be fundamental functions in engineering, and it is strongly recommended that the reader becomes very familiar with both of these functions.

$$u(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t > 0 \\ \frac{1}{2} & \text{if } t = 0 \end{cases}$$

Sinc signal: There is a particular form that appears so frequently in communications engineering, that we give it its own name. This function is called the "Sinc function".

The Sinc function is defined in the following manner:

$$\text{sinc}(x) = \frac{\sin \pi x}{\pi x} \quad \text{if } x \neq 0 \quad \text{and} \quad \text{sinc}(0) = 1$$

The value of sinc(x) is defined as 1 at x = 0, since

$$\lim_{x \rightarrow 0} \text{sinc}(x) = 1$$

PROCEDURE:-

- Open MATLAB
- Open new M-file
- Type the program
- Save in current directory
- Compile and Run the program
- For the output see command window\ Figure window

PROGRAM:

% Generation of signals and sequences

clc;

clear all;

close all;

%~~~~~

%generation of unit impulse signal

t1=-1:0.01:1

y1=(t1==0);

subplot(2,2,1);

plot(t1,y1);

xlabel('time');

ylabel('amplitude');

title('unit impulse signal');

%generation of impulse sequence

subplot(2,2,2);

stem(t1,y1);

xlabel('n');

ylabel('amplitude');

title('unit impulse sequence');

%~~~~~

%generation of unit step signal

t2=-10:1:10;

y2=(t2>=0);

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subplot(2,2,3);
plot(t2,y2);
xlabel('time');
ylabel('amplitude');
title('unit step signal');
%generation of unit step sequence
subplot(2,2,4);
stem(t2,y2);
xlabel('n');
ylabel('amplitude');
title('unit step sequence');
%~~~~~

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```

%generation of square wave signal
t=0:0.002:0.1;
y3=square(2*pi*50*t);
figure; subplot(2,2,1);
plot(t,y3);
axis([0 0.1 -2 2]);
xlabel('time');
ylabel('amplitude');
title('square wave signal');
%generation of square wave sequencesubplot(2,2,2);
stem(t,y3);
axis([0 0.1 -2 2]);
xlabel('n');
ylabel('amplitude');
title('square wave sequence');
%~~~~~

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%generation of sawtooth signal
y4=sawtooth(2*pi*50*t);
subplot(2,2,3);
plot(t,y4);
axis([0 0.1 -2 2]);
xlabel('time');
ylabel('amplitude');
title('sawtooth wave signal');
%generation of sawtooth sequence
subplot(2,2,4);
stem(t,y4);

```

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axis([0 0.1 -2 2]);
xlabel('n');
ylabel('amplitude');
title('sawtooth wave sequence');
%~~~~~

%generation of triangular wave signal
y5=sawtooth(2*pi*50*t,.5);
figure; subplot(2,2,1);
plot(t,y5);
axis([0 0.1 -2 2]);
xlabel('time');
ylabel('amplitude');
title('triangular wave signal');
%generation of triangular wave sequence
subplot(2,2,2);
stem(t,y5);
axis([0 0.1 -2 2]);
xlabel('n');
ylabel('amplitude');
title('triangular wave sequence');
%~~~~~

%generation of sinusoidal wave signal
y6=sin(2*pi*40*t);
subplot(2,2,3);
plot(t,y6);
axis([0 0.1 -2 2]);
xlabel('time');
ylabel('amplitude');
title('sinusoidal wave signal');
%generation of sin wave sequence
subplot(2,2,4);
stem(t,y6);
axis([0 0.1 -2 2]);
xlabel('n');
ylabel('amplitude');
title('sin wave sequence');
%~~~~~

%generation of ramp signal
y7=t;

```

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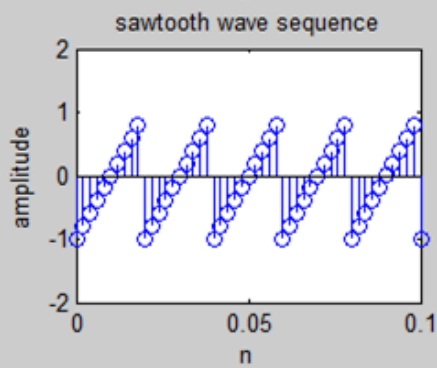
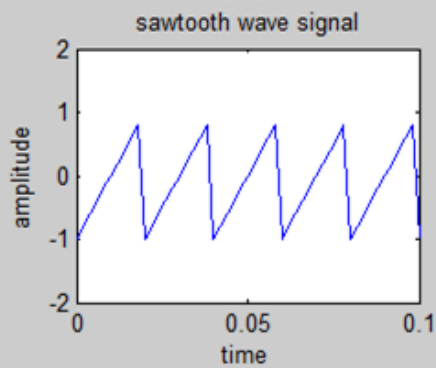
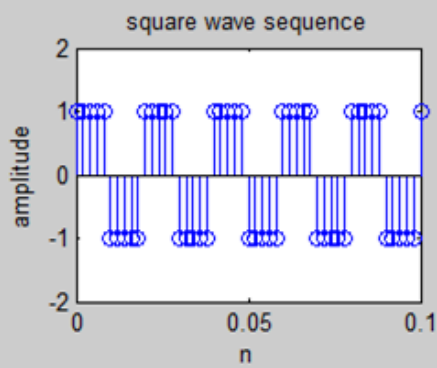
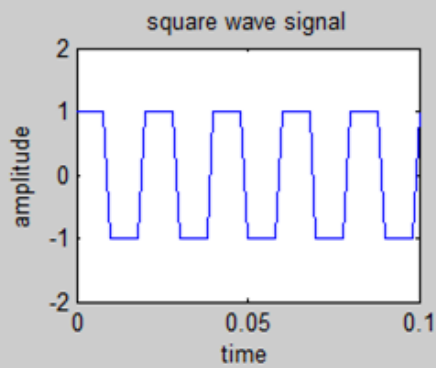
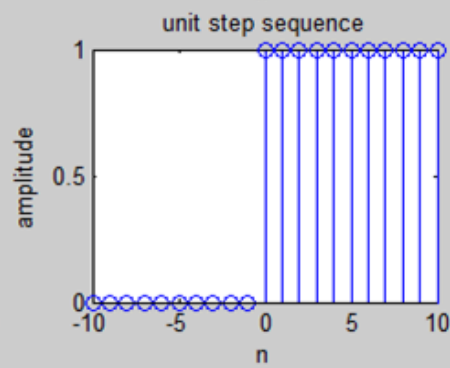
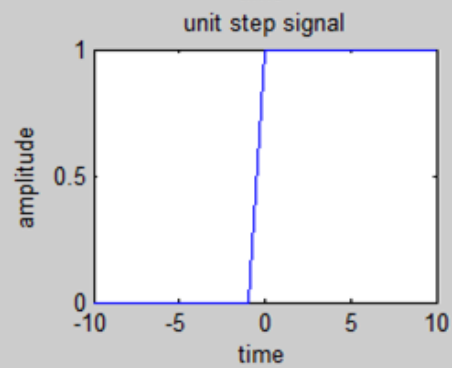
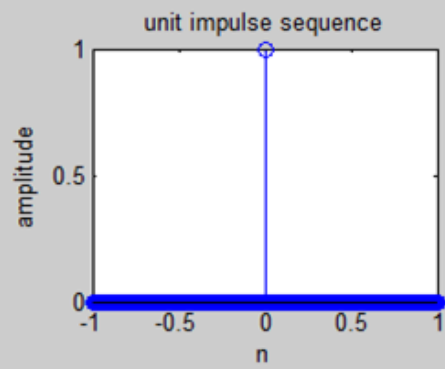
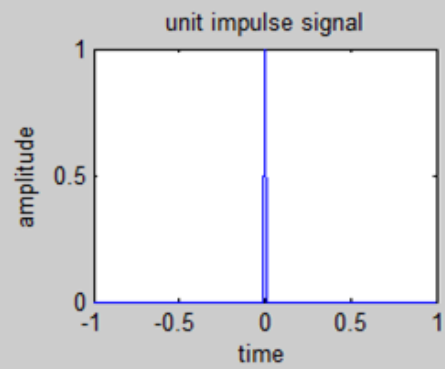
figure;
subplot(2,2,1);
plot(t,y7);
xlabel('time');
ylabel('amplitude');
title('ramp signal');
%generation of ramp sequence
subplot(2,2,2);
stem(t,y7);
xlabel('n');
ylabel('amplitude');
title('ramp sequence');
%~~~~~

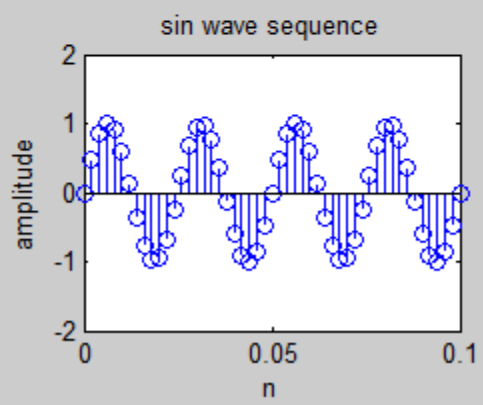
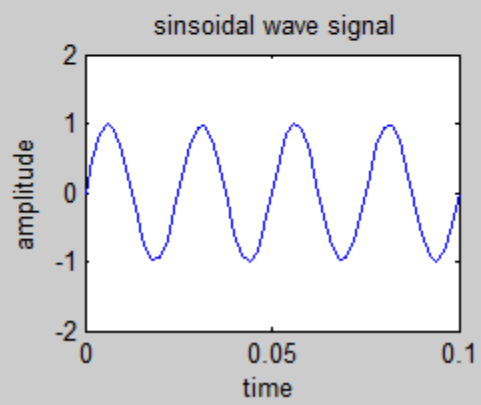
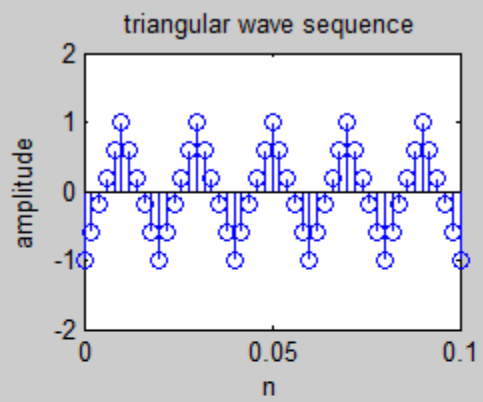
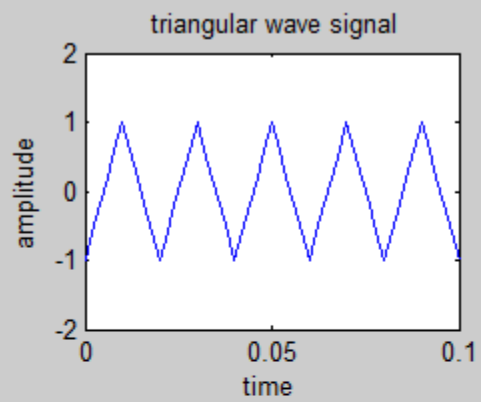
%generation of sinc signal
t3=linspace(-5,5);
y8=sinc(t3);
subplot(2,2,3);
plot(t3,y8);
xlabel('time');
ylabel('amplitude');
title('sinc signal');
%generation of sinc sequence
subplot(2,2,4);
stem(y8); xlabel('n');
ylabel('amplitude');
title('sinc sequence');

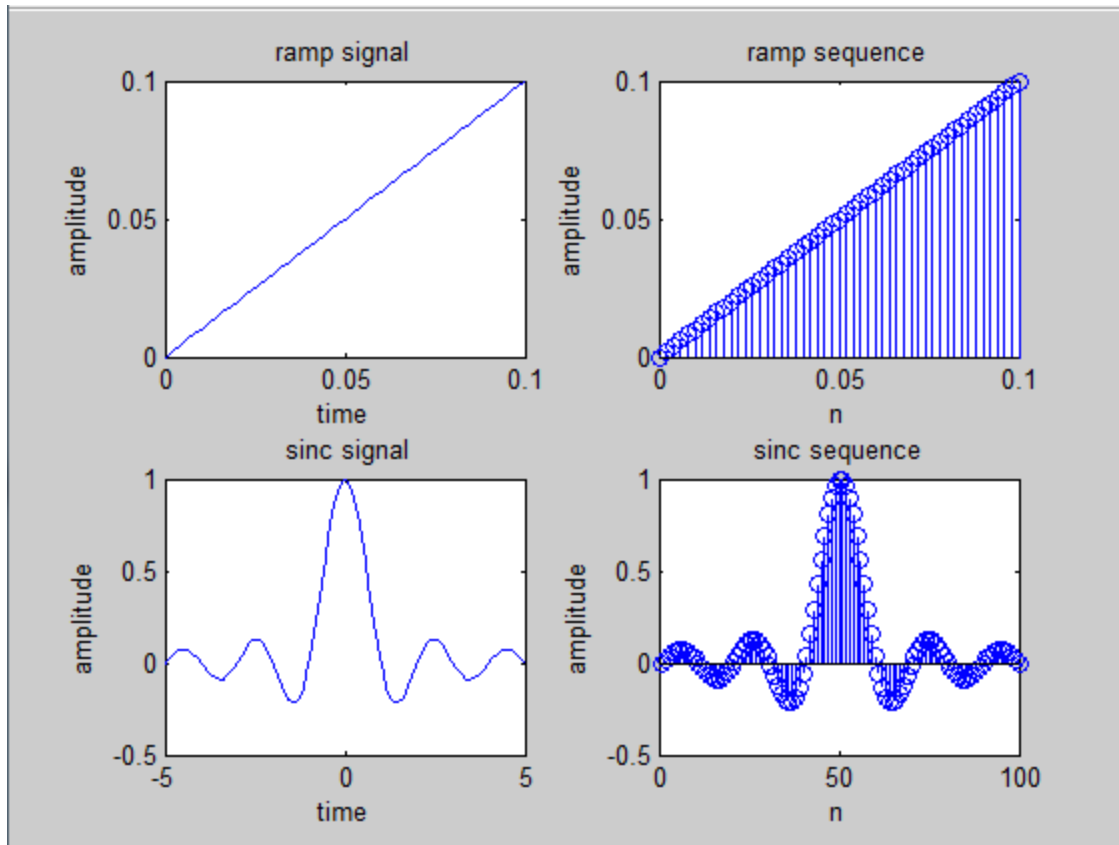
```

Result: Various signals & sequences generated using Matlab software.

output:







VIVA QUESTIONS:-

1. Define Signal?
2. Define continuous and discrete Signals?
3. State the relation between step, ramp and Delta Functions?
4. Differentiate saw tooth and triangular signals?
5. Define Periodic and aperiodic Signal?